OnTrack Style Guide v2.0 (Draft)

Implementation Framework

Executive Summary

This comprehensive style guide establishes OnTrack as a modern, accessible, and maintainable learning management system through enterprise-grade CSS architecture, WCAG 2.2 compliance, and scalable design systems implementation.

Key Achievements

- Complete accessibility audit and remediation strategy
- Enterprise CSS architecture aligned with Google's technical standards
- Modern design token system with multi-theme support
- Automated testing and governance framework
- Professional documentation and implementation guidelines

1. Project Architecture Analysis

1.1. Current Technology Stack

OnTrack operates as a hybrid Angular application currently migrating from AngularJS/CoffeeScript to Angular 17+/TypeScript. The analysis reveals -

Modern Stack

- Angular 17.3.6 with Angular Material design system
- TypeScript 5.2 for type safety and maintainability
- Tailwind CSS 3.3 for utility-first styling
- SCSS/Sass for component styling
- Angular CLI with modern build processes

Legacy Stack (Being Migrated)

- AngularJS 1.5.11 with CoffeeScript
- Bootstrap 3.4 framework
- Grunt build system with LESS preprocessing

1.2. Existing Styling Architecture

The current implementation demonstrates a sophisticated but improvable styling system -

```
// Current theme structure (src/theme.scss)
$md-formatif: (
500: #3939ff, // Primary brand color
// ... comprehensive Material Design palette
);
// Task status colors (src/styles/common/task-status-colors.scss)
$task-status-colors: (
complete: (base: #5bb75b, fore: #fff),
ready-for-feedback: (base: #0079d8, fore: #fff),
// ... 12 distinct status indicators
);
```

Strengths Identified

- Comprehensive task status color system with accessibility considerations
- Material Design integration with custom theming
- Component-scoped styling approach
- Systematic color palette management

Critical Gaps

- Lack of centralized design token system
- No automated accessibility testing
- Limited color blindness support
- Inconsistent CSS architecture patterns

2. Enhanced CSS Architecture Framework

2.1. ITCSS Implementation Strategy

Based on research into scalable CSS methodologies, OnTrack should implement Inverted Triangle CSS (ITCSS) architecture -

```
styles/
├── 01-settings/ # Design tokens, variables
│  ├── _design-tokens.scss
│  ├── _colors.scss
│  └── _typography.scss
├── 02-tools/ # Mixins, functions
│  ├── _accessibility.scss
│  └── _layout-helpers.scss
├── 03-generic/ # Reset, normalize
├── 04-elements/ # Base HTML styling
├── 05-objects/ # Layout patterns
├── 06-components/ # UI components
│  ├── _task-status.scss
│  ├── _navigation.scss
│  └── _burndown-chart.scss
└── 07-utilities/ # Helper classes
```

2.2. Enhanced BEM Methodology

Building on the existing BEM approach, the enhanced naming convention provides improved clarity -

```
// Enhanced BEM with accessibility context
.task-status {
// Base component
display: inline-flex;
min-height: 24px; // WCAG 2.2 target size requirement
&amp;__icon {
// Element: visual indicator
width: 16px;
```

```
height: 16px;
// Accessibility: shape differentiation
&amp;amp;--complete { border-radius: 50%; } // Circle
&amp;amp;--overdue { clip-path: polygon(...); } // Star shape
&amp;amp;--pending { border-radius: 2px; } // Square
}
&amp;amp;__label {
// Element: text description
font-weight: 500;
margin-left: 8px;
}
// Modifiers: status variants with accessibility
&amp;amp;--complete {
background-color: var(--color-success-100);
color: var(--color-success-700);
border: 1px solid var(--color-success-300);
}
}
```

2.3. Design Token Architecture

Implementation of a comprehensive design token system -

```
// 01-settings/_design-tokens.scss
:root {
// Color tokens - WCAG 2.2 compliant
--color-primary-50: #e7e7ff;
--color-primary-500: #3939ff;
2.2 Enhanced BEM Methodology
2.3 Design Token Architecture
--color-primary-900: #1717ff;
// Semantic tokens
--color-success-base: #2e8540;
--color-error-base: #b00020;
--color-warning-base: #c56a00;
// Typography tokens
--font-family-primary: 'Grotesk';
--font-family-body: 'Inter', 'Roboto', sans-serif;
--font-weight-normal: 400;
--font-weight-semibold: 600;
// Spacing tokens
--space-xs: 0.25rem;
--space-sm: 0.5rem;
```

```
--space-md: 1rem;
--space-lg: 1.5rem;
--space-xl: 2rem;
// Accessibility tokens
--focus-ring-width: 2px;
--focus-ring-color: var(--color-primary-500);
--target-size-minimum: 24px;
}
```

3. Accessibility Implementation

3.1. WCAG 2.2 Compliance Framework

Complete implementation addressing all new WCAG 2.2 success criteria -

New Requirements Implementation -

- **2.4.11 Focus Not Obscured** - Focus indicators remain visible with 2px outline
- **2.4.12 Focus Not Obscured (Enhanced)** - No UI elements obscure focused elements
- **2.5.7 Dragging Movements** - Alternative single-click actions for all drag operations
- **2.5.8 Target Size (Minimum)** - 24×24px minimum for all interactive elements
- **3.2.6 Consistent Help** - Help mechanisms in consistent locations
- **3.3.7 Redundant Entry** - Form data persistence and auto-completion
- **3.3.8 Accessible Authentication** - No cognitive function tests required

3.2. Color Blindness Support System

```
// Enhanced color system with shape and pattern support
.task-status {
// Base accessibility requirements
position: relative;
&amp;::before {
// Pattern overlay for color blind users
```

```
content: '';
position: absolute;
top: 0;
left: 0;
right: 0;
bottom: 0;
pointer-events: none;
}
&amp;--complete::before {
background-image: url('data:image/svg+xml,&lt;svg&gt;...&lt;/svg&gt;');
// Checkmark
}
&amp;--overdue::before {
background-image: url('data:image/svg+xml,&lt;svg&gt;...&lt;/svg&gt;');
// Warning pa
}
&amp;--pending::before {
background-image: url('data:image/svg+xml,&lt;svg&gt;...&lt;/svg&gt;');
// Pending pa
}
}
// Contrast-safe color palette
$accessible-colors: (
success: (
base: #2e8540, // 4.5:1 contrast on white
alternative: #1b7332, // High contrast alternative
),
error: (
base: #b00020, // 4.5:1 contrast on white
alternative: #8b0000, // High contrast alternative
),
warning: (
base: #c56a00, // 4.5:1 contrast on white
alternative: #a0550a, // High contrast alternative
)
);
```

3.3. Automated Testing Integration

```
// accessibility-testing.js - Integrated into CI/CD pipeline
import { axeCheck } from '@axe-core/playwright';
import colorBlind from 'color-blind';
export async function validateAccessibility(page, component) {
```

```
// Automated axe testing
const results = await axeCheck(page, component);
// Color blindness simulation testing
const elements = await page.$$(component);
for (const element of elements) {
const styles = await element.computedStyle();
// Test multiple color blindness types
const simulations = ['deuteranopia', 'protanopia', 'tritanopia'];
for (const type of simulations) {
const simulatedBg = colorBlind[type](styles.backgroundColor);
const simulatedText = colorBlind[type](styles.color);
const contrast = calculateContrast(simulatedBg, simulatedText);
if (contrast &lt; 4.5) {
console.error(`Insufficient contrast for ${type}: ${contrast}`);
}
}
}
return results;
}
```

4. Component Implementation Examples

4.1. Enhanced Task Status Component

```
// 06-components/_task-status.scss
.task-status {
// Accessibility-first design
display: inline-flex;
align-items: center;
gap: var(--space-xs);
padding: var(--space-xs) var(--space-sm);
border-radius: 4px;
// WCAG 2.2 requirements
min-height: var(--target-size-minimum);
min-width: var(--target-size-minimum);
// Focus management
&amp;:focus-visible {
outline: var(--focus-ring-width) solid var(--focus-ring-color);
outline-offset: 2px;
}
// Icon with semantic meaning
&amp;__icon {
```

```
width: 16px;
height: 16px;
flex-shrink: 0;
// Shape differentiation for accessibility
&amp;;--complete {
background: var(--color-success-base);
border-radius: 50%;
position: relative;
&amp;::after {
content: '✓';
color: white;
position: absolute;
top: 50%;
left: 50%;
transform: translate(-50%, -50%);
font-size: 10px;
}
}
&amp;;--overdue {
background: var(--color-error-base);
clip-path: polygon(50% 0%, 61% 35%, 98% 35%, 68% 57%, 79% 91%, 50% 70%,
21% 91%, 32
}
&amp;;--pending {
background: var(--color-neutral-400);
border-radius: 2px;
border: 2px solid var(--color-neutral-600);
}
}
// Status-specific styling with high contrast
&amp;;--complete {
background-color: color-mix(in srgb, var(--color-success-base) 10%,
transparent);
color: var(--color-success-900);
border: 1px solid var(--color-success-300);
}
&amp;;--overdue {
background-color: color-mix(in srgb, var(--color-error-base) 10%,
transparent);
color: var(--color-error-900);
border: 1px solid var(--color-error-300);
}
}
// Dark theme support
[data-theme="dark"] {
.task-status {
```

```
&amp;--complete {
background-color: color-mix(in srgb, var(--color-success-base) 20%,
transparent);
color: var(--color-success-100);
border-color: var(--color-success-600);
}
}
}
// High contrast theme
[data-theme="high-contrast"] {
.task-status {
border-width: 2px;
&amp;--complete {
background-color: var(--color-success-base);
color: white;
border-color: white;
}
}
}
```

4.2. Accessible Burndown Chart Implementation

```
// 06-components/_burndown-chart.scss
.burndown-chart {
// Container styling
position: relative;
background: var(--color-surface-primary);
border-radius: 8px;
padding: var(--space-md);
// Accessibility announcement
&amp;[aria-describedby] {
// Connected to screen reader description
}
// Chart lines with accessibility patterns
.chart-line {
stroke-width: 3px; // Increased for visibility
&amp;--target {
stroke: var(--color-neutral-700);
stroke-dasharray: none; // Solid line
}
&amp;--actual {
stroke: var(--color-primary-600);
stroke-dasharray: 5,5; // Dashed pattern
```

```
}
&amp;amp;--projected {
stroke: var(--color-warning-600);
stroke-dasharray: 10,5,2,5; // Dot-dash pattern
}
}
// Data markers for pattern recognition
.chart-marker {
&amp;amp;--target {
fill: var(--color-neutral-700);
r: 4px;
shape-rendering: crispEdges; // Square markers
}
&amp;amp;--actual {
fill: var(--color-primary-600);
r: 3px; // Circle markers (default)
}
&amp;amp;--projected {
fill: var(--color-warning-600);
r: 4px;
transform: rotate(45deg); // Diamond markers
}
}
// Accessible legend
.chart-legend {
display: flex;
gap: var(--space-md);
margin-top: var(--space-sm);
&amp;amp;__item {
display: flex;
align-items: center;
gap: var(--space-xs);
font-size: 14px;
&amp;amp;-marker {
width: 24px;
height: 3px;
border-radius: 1px;
}
}
}
}
// Screen reader accessible description
.chart-description {
position: absolute;
width: 1px;
height: 1px;
```

```
padding: 0;
margin: -1px;
overflow: hidden;
clip: rect(0, 0, 0, 0);
white-space: nowrap;
border: 0;
}
```

5. Theme System Implementation

5.1. Multi-Theme Architecture

```
// Theme configuration with CSS custom properties
:root {
// Default (light) theme
--theme-surface-primary: #ffffff;
--theme-surface-secondary: #f8f9fa;
--theme-text-primary: #2c2c2c;
--theme-text-secondary: #666666;
}
[data-theme="dark"] {
--theme-surface-primary: #121212;
--theme-surface-secondary: #1e1e1e;
--theme-text-primary: #ffffff;
--theme-text-secondary: #b3b3b3;
// Adjusted contrast ratios for dark mode
--color-success-base: #81c784;
--color-error-base: #e57373;
--color-warning-base: #ffb74d;
}
[data-theme="high-contrast"] {
--theme-surface-primary: #000000;
--theme-surface-secondary: #000000;
--theme-text-primary: #ffffff;
--theme-text-secondary: #ffffff;
// Maximum contrast colors
--color-success-base: #00ff00;
```

```
--color-error-base: #ff0000;
--color-warning-base: #ffff00;
}
[data-theme="color-blind-safe"] {
// Optimized for color vision deficiencies
--color-primary-base: #1f77b4; // Blue (safe for all types)
--color-secondary-base: #ff7f0e; // Orange (distinguishable)
--color-success-base: #2ca02c; // Green (adjusted luminance)
--color-error-base: #d62728; // Red (distinct luminance)
--color-warning-base: #9467bd; // Purple (alternative to yellow)
}
```

5.2. Dynamic Theme Switching

```
// Theme service implementation
@Injectable({
providedIn: 'root'
})
export class ThemeService {
private currentTheme = 'default';
setTheme(theme: 'default' | 'dark' | 'high-contrast' |
'color-blind-safe') {
document.documentElement.setAttribute('data-theme', theme);
this.currentTheme = theme;
localStorage.setItem('ontrack-theme', theme);
}
initializeTheme() {
const savedTheme = localStorage.getItem('ontrack-theme');
const prefersDark = window.matchMedia('(prefers-color-scheme:
dark)').matches;
const prefersHighContrast = window.matchMedia('(prefers-contrast:
high)').matches;
if (savedTheme) {
this.setTheme(savedTheme as any);
} else if (prefersHighContrast) {
this.setTheme('high-contrast');
} else if (prefersDark) {
this.setTheme('dark');
}
}
```

```
}
```

6. Build Process and Automation

6.1. PostCSS Configuration

```
// postcss.config.js - Enhanced build pipeline
module.exports = {
plugins: [
require('postcss-import'),
require('postcss-custom-properties'),
// Accessibility testing during build
require('postcss-colorblind')({
method: 'deuteranopia'
}),
// Design token processing
require('postcss-design-tokens'),
// Optimization
require('autoprefixer'),
require('cssnano')({
preset: ['default', {
discardComments: { removeAll: true }
}]
})
]
};
```

6.2. Automated Testing Pipeline

```
# .github/workflows/accessibility-testing.yml
name: Accessibility Testing
on: [push, pull_request]
jobs:
accessibility:
runs-on: ubuntu-latest
steps:
- uses: actions/checkout@v3
- name: Setup Node.js
uses: actions/setup-node@v3
with:
node-version: '20'
```

```
- name: Install dependencies
run: npm ci
- name: Run accessibility tests
run: |
npm run test:accessibility
npm run test:color-blindness
npm run lint:styles
- name: Generate accessibility report
run: npm run report:accessibility
- name: Upload reports
uses: actions/upload-artifact@v3
with:
name: accessibility-reports
path: reports/
```

7. Documentation and Governance

7.1. Living Style Guide Implementation

```
// Storybook configuration for component documentation
module.exports = {
stories: ['../src/**/*.stories.@(js|jsx|ts|tsx)'],
addons: [
'@storybook/addon-essentials',
'@storybook/addon-accessibility',
'@storybook/addon-design-tokens'
],
features: {
buildStoriesJson: true
},
// Accessibility testing integration
previewAnnotations: [
require.resolve('./accessibility-preview.js')
]
};
```

7.2. Development Guidelines

CSS Authoring Rules -

I. Always use design tokens instead of hardcoded values
II. Include accessibility considerations in every component
III. Test with color blindness simulators before committing
IV. Validate WCAG 2.2 compliance using automated tools
V. Document component variations and accessibility features

Code Review Checklist -

- Design tokens used consistently
- WCAG 2.2 compliance verified
- Color blindness alternatives provided
- Focus management implemented
- Screen reader compatibility tested
- Cross-browser testing completed

8. Implementation Roadmap

8.1. Phase 1: Foundation

I. Design Token Migration
   A. Convert existing color variables to design tokens
   B. Implement ITCSS directory structure
   C. Setup automated build pipeline
II. Accessibility Infrastructure
   A. Integrate axe-core testing
   B. Setup color blindness simulation tools
   C. Create accessibility testing documentation

8.2. Phase 2: Component Enhancement

I. Task Status System Overhaul
   A. Implement shape-based differentiation
   B. Add pattern overlays for color blind users

C. Create comprehensive status documentation

II. Chart Accessibility

A. Add line pattern differentiation

B. Implement data markers and legends

C. Create screen reader alternatives

8.3. Phase 3: Advanced Features

I. Multi-Theme Implementation

A. Dark mode with proper contrast ratios

B. High contrast theme for low vision users

C. Color blind safe theme variations

II. Automated Governance

A. CI/CD integration for accessibility testing

B. Style guide generation and deployment

C. Performance monitoring and optimization

8.4. Phase 4: Documentation and Training

I. Comprehensive Documentation

A. Living style guide with Storybook

B. Developer guidelines and best practices

C. Accessibility testing procedures

II. Team Training and Adoption

A. Developer workshops on new architecture

B. Design team collaboration processes

C. Quality assurance integration

9. Success Metrics and Validation

9.1. Accessibility Compliance Metrics

- **WCAG 2.2 AA Compliance -** 100% automated test pass rate

- **Color Contrast Ratios** - Minimum 4.5:1 for normal text, 3:1 for large text
- **Color Independence** - All status information conveyed through multiple channels
- **Keyboard Navigation** - Complete functionality without mouse interaction

9.2. Performance Benchmarks

- **CSS Bundle Size** - <50KB gzipped for production build
- **First Contentful Paint** - 1.5 seconds on 3G networks
- **Cumulative Layout Shift** - <0.1 score
- **Theme Switching** - <100ms transition time

9.3. Developer Experience Metrics

- **Build Time** - <30 seconds for style compilation
- **Code Review Time** - <15 minutes for style-related changes
- **Onboarding Time** - <2 hours for new developers to understand system
- **Documentation Coverage** - 100% of components documented in Storybook